AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 11 · STUDENT LESSON

Failure budgets and service objectives

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Define success. Count requests once. Preserve the window.

Evaluate a count-based service objective with explicit eligibility,integer failure allowance and unchanged original history.

Keep one habit: Objective contract → complete count ledger → separate counterfactual picture → overlap/eligibility probes → executable audit → independent objective.

MAIN LESSON · SECTION 1

Define success before counting the failure budget

> A reliability percentage needs an interval, eligible requests and a fixed contract.

Learning goals

I can explain the **mechanism** and terminology of **failure budget**s and **service objective**s.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve evaluation coverage and preserved outcomes

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Evaluation pipelines. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Authentication and authorisation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Evaluation pipelines. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain **failure budget**s and **service objective**s using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of **failure budget**s and **service objective**s

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Write the objective as an observable contract

> Eligibility and success rules must not change to hide a missed target.

Begin with the school-service target

A school service can set a target such as "at least 98% of eligible requests pass all agreed checks in this observation window." That is a rule to judge a particular record, not a promise that nothing ever fails. Define which requests belong in the window and what counts as success before inspecting outcomes.

A correct, timely refusal can be a success when refusal is the required behaviour; an incorrect answer cannot become a success merely because the service returned text.

Define service objectives and failure budgets

Service objectives define measured reliability over a stated interval. A **failure budget** is meaningful only after success and **eligible request**s are defined.

Complete window,scope and counting policy

Declared objective contract	Complete local exercise
Observation window	Fictional W-A, 09:00–10:00 on one fictional day; supplied records assigned to that window
Eligibility	200 user-request fixtures; 3 separate internal health checks excluded by predeclared policy
Success	All supplied schema, support-or-correct-refusal and timing-contract flags pass
Target	At least 98% success over eligible requests
Count rule	A request with several failed checks counts as one failed request
Maximum failures	floor((100–target percent) × eligible /100) for integer-percent target
Actual evidence	Constructed check flags and count arithmetic; no real requests or timing measured
Report	Success/failure counts, eligible/excluded counts, target, budget and interval identity

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Compute the budget and compare the actual count

> Keep percentage points,request counts and eligible denominator distinct.

Original objective and allowed-failure derivation

For a fictional objective at least 98% of 200 **eligible request**s must pass all contract checks. Allowed failures are 2%*200=4. Count every eligible failure once; report excluded requests separately.

Complete count-to-decision ledger

Main W-A count	Calculation	Result
Eligible requests	200	Fixed denominator
Failed requests	5 distinct request IDs	Count once per request
Successes	200−5	195
Observed fixture success rate	195/200	97.5%
Maximum integer failures at 98%	(100−98)×200/100	4
Budget remaining	4−5	−1: exceeds allowance by1
Objective decision	97.5% <98%	Does not meet this window’s target

Main W-A count	Calculation	Result
Excluded health checks	3 separate records	Report outside eligible denominator

State the decision boundary before judging

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

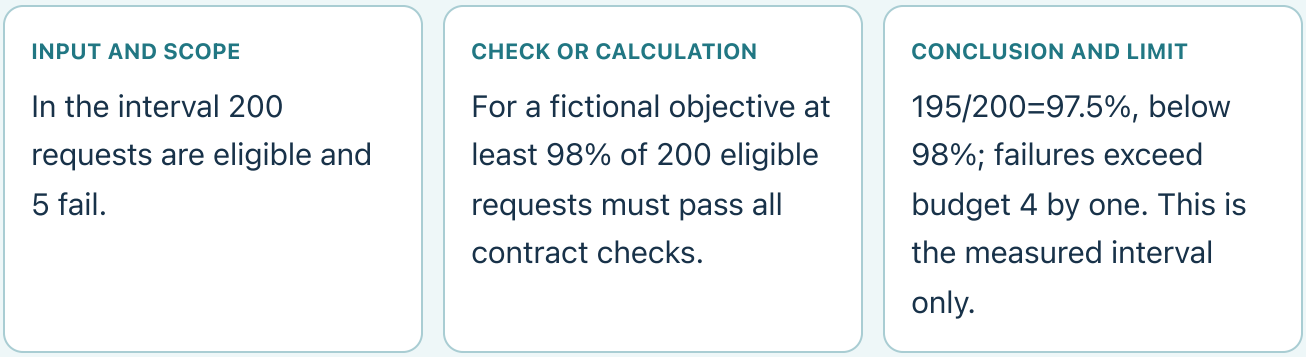
If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Failure budgets and service objectives — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change a hypothetical window without rewriting the original

> A counterfactual example and a real retry have different evidence roles.

Current task and response

- In the interval 200 requests are eligible and 5 fail. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the separate pictured counterfactual

- Change: Counterfactual: change one failed outcome to a success in a separate copy of this 100-request window.

▼ **Check after predicting**

Prediction: 98 successes and 2 failures meet the stated window target.

Reason: The counterfactual copy changes one outcome while its denominator stays 100; preserve the original 97-success/3-failure window.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Keep the original window and its hypothetical variation side by side

> Meeting one interval’s aggregate target does not establish future reliability.

Main supplied count and conclusion

In the interval 200 requests are eligible and 5 fail.

195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only.

Keep the reasoning’s scope

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate100-request picture and success definition

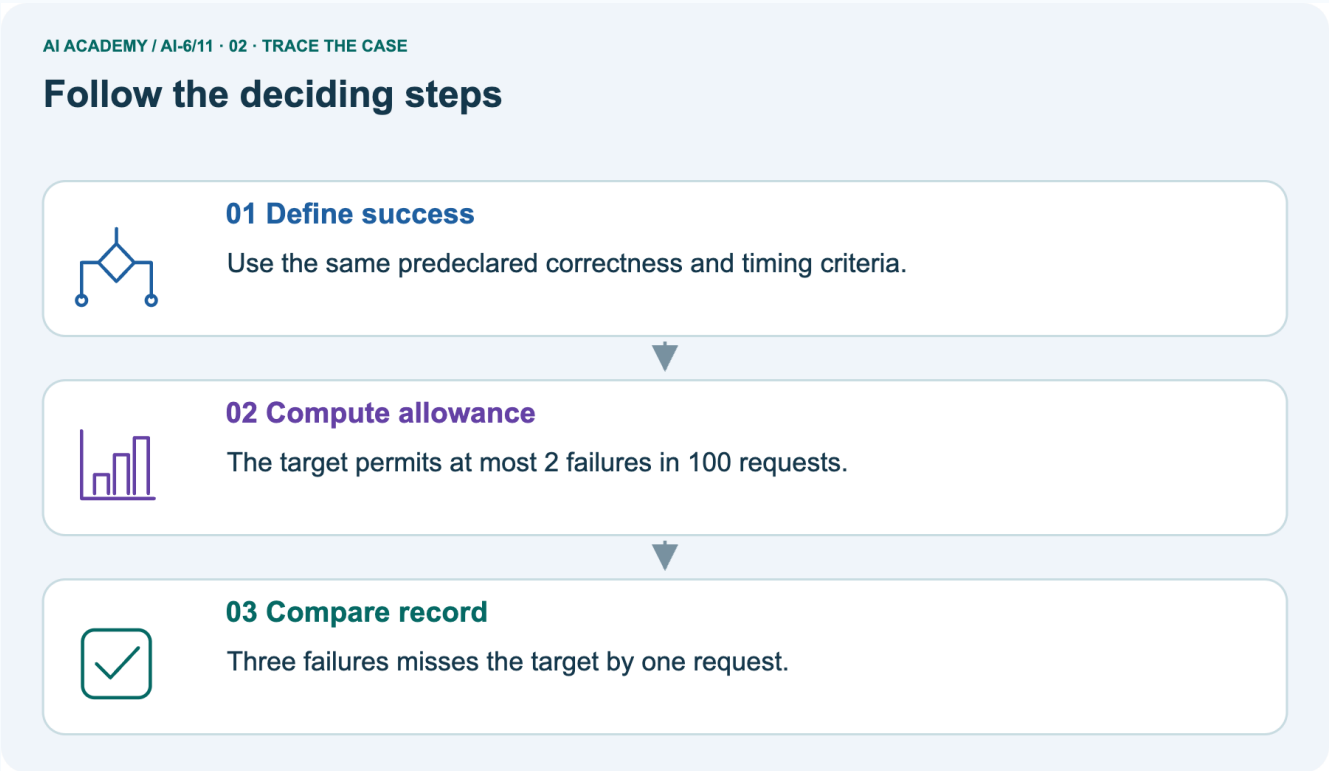
A reliability allowance. A practice service requires at least 98 successes in each window of 100 **eligible request**s.

Current window has 97 successes and 3 failures. Success includes a correct refusal within the agreed time limit.

The main window has 200 eligible requests and 5 failures. The picture is a separate 100-request window:97 successes, 3 failures, target at least 98 successes. Its maximum allowance is2 failures, so it misses by1. The changed picture is a counterfactual copy with 98 successes and 2 failures, which meets the same rule. Preserve the original 97/3 history.

In a real service, a later successful retry does not automatically erase the original failed request; request identity, retry counting and window policy must be declared before measurement.

Trace allowed count and missed target



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Define success	Use the same predeclared correctness and timing criteria.
2. Compute allowance	The target permits at most 2 failures in 100 requests.
3. Compare record	Three failures misses the target by one request.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An objective needs a measurement window, denominator and success definition. An honest refusal can be success under an agreed contract.

Predict the explicitly separate counterfactual

- Change one thing: Counterfactual: change one failed outcome to a success in a separate copy of this 100-request window. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting

AI ACADEMY / AI-6/11 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

Three failures misses the target by one request.

Changed condition

Counterfactual: change one failed outcome to a success in a separate copy of this 100-request window.

Worked changed-case result

98 successes and 2 failures meet the stated window target.
Why: The counterfactual copy changes one outcome while its denominator stays 100; preserve the original 97-success/3-failure window.

Follow the labelled picture with the matching explanation. Open the vector diagram

Separate pictured record · Successes / eligible · Failures · Allowed failures · Decision

> **Separate pictured record:** Original window, retain unchanged
>
> **Successes / eligible:** 97/100
>
> **Failures:** 3
>
> **Allowed failures:** 2
>
> **Decision:** Misses target

> **Separate pictured record:** Counterfactual copy, one different outcome
>
> **Successes / eligible:** 98/100
>
> **Failures:** 2
>
> **Allowed failures:** 2

> **Decision:** Meets target at equality

> **Separate pictured record:** Actual retry after an observed failure
>
> **Successes / eligible:** Requires declared request/window policy
>
> **Failures:** Do not silently erase prior failure
>
> **Allowed failures:** Keep original history
>
> **Decision:** Not supplied or measured here

Explain equality,history and remaining limits

- Meeting one window does not guarantee the next; monitor repeated windows and distinguish failure types.

- Read the labelled picture: Calculate the maximum allowed failures among 100 requests. Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?

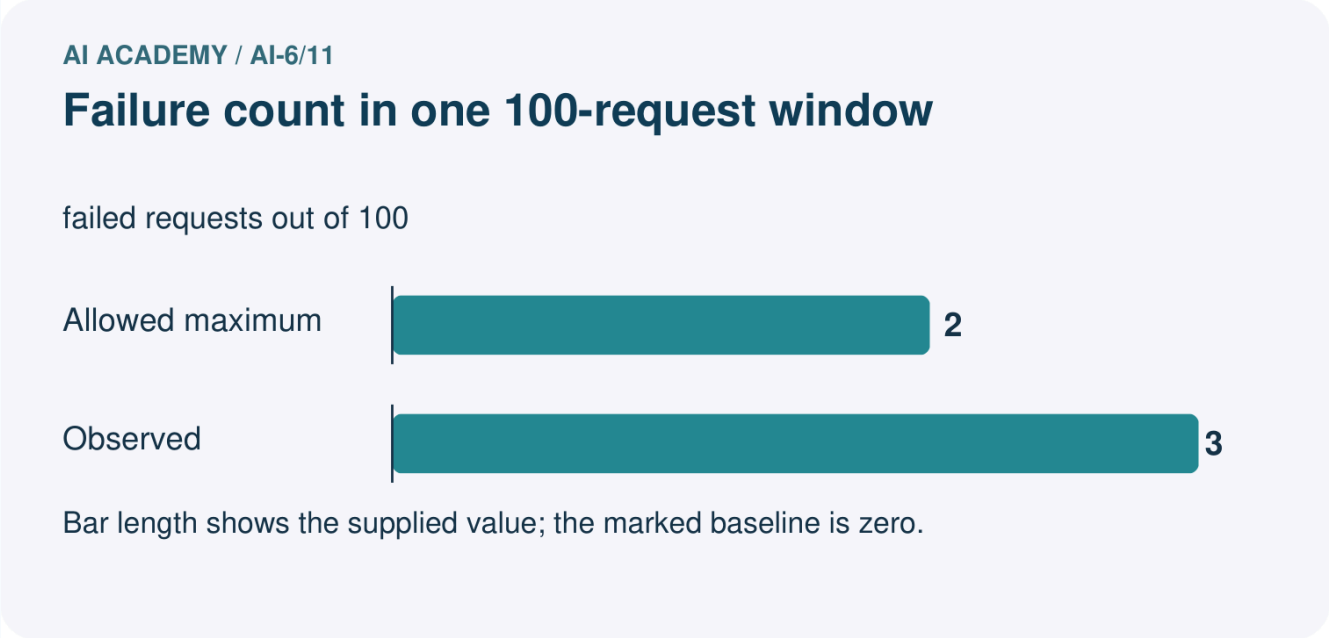
<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.
</details>

A pictorial worked case: Failure budgets and service objectives

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-6/11

Failure budgets and service objectives

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Define success

Use the same predeclared correctness and timing criteria.



2. Compute allowance

The target permits at most 2 failures in 100 requests.



3. Compare record

Three failures misses the target by one request.

CHANGE ONE THING

Repair one failed request into a success in this window.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Audit overlap,eligibility and integer boundaries

> A failure budget is an accounting rule,not permission to ignore critical failures.

Core one-interval qualification

Meeting one interval objective does not establish every future interval or zero critical failures.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete counting and evidence-limit probes

Budget or evidence probe	Correct interpretation
One request fails support and schema	One failed request, two failed criteria; do not double-count the request
Correct refusal passes every required check	Can count as success under the declared contract
Three internal health checks excluded by predeclared rule	Report separately; do not remove failing eligible user requests post hoc
98% target over 12 requests	At most 0 integer failures; do not round 0.24 up to 1
No eligible requests	Empirical success rate unavailable; this helper rejects zero denominator
Aggregate target met with a severe incident	Does not prove zero critical failures or make the incident acceptable
A later retry succeeds	Retain original history and apply the predeclared retry/window rule

Budget or evidence probe	Correct interpretation
One constructed window meets or misses target	No guarantee about other windows or future populations

Guided explanation and repair

- A report omits this qualification: Meeting one interval objective does not establish every future interval or zero critical failures.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise objective decisions with fixed definitions

> Name the interval,eligible count,success rule and allowed failures.

Current explanation and guided scope check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 record

Use workbook W2 for the connected example "**Failure budget**s and **service objective**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual future-window limit check

- Use the worked case for **Failure budget**s and **service objective**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured budget response

- Calculate the maximum allowed failures among 100 requests. Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?

> **My explanation using a pictured step and the given facts**

Reserve independent 95%/80-request case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete count-based objective and overlap audit

> Keep request failures,criterion failures and excluded events distinct.

Current 200-request objective investigation

In the interval 200 requests are eligible and 5 fail. Show the intermediate steps and give the conclusion with its scope.

Complete runnable fixtures and exact threshold helper

```
"""Exact count-based objectives on constructed request-check fixtures."""
def objective(eligible,failures,target_percent):
    if (type(eligible) is not int or eligible<=0 or type(failures) is not int
        or not 0<=failures<=eligible or type(target_percent) is not int
        or not 0<=target_percent<=100):
        raise ValueError('Need positive eligible count,valid failures,integer percent')
    successes=eligible-failures
    allowed_failures=((100-target_percent)*eligible)//100
    return {'eligible':eligible,'successes':successes,'failures':failures,
            'allowed_failures':allowed_failures,'success_rate':successes/eligible,
            'meets':100*successes>=target_percent*eligible,
            'remaining_budget':allowed_failures-failures}

WINDOW={'id':'W-A','start':'fictional day 09:00','end':'fictional day 10:00',
        'scope':'supplied user-request ledger assigned to this window'}
requests=[]
for i in range(1,201):
    checks={'schema':True,'support_or_correct_refusal':True,'time_contract':True}
    if i in [196,199,200]:checks['support_or_correct_refusal']=False
    if i in [197,199]:checks['schema']=False
    if i in [198,200]:checks['time_contract']=False
    requests.append({'id':f'R{i:03}','eligible':True,
                     'kind':'correct_refusal' if i==10 else 'answer', 'checks':checks})
# Predeclared separate health-check events are not user requests in this toy policy.
excluded=[{'id':f'H{i}','eligible':False,'reason':'internal health check'} for i in
range(1,4)]
failed=[r for r in requests if not all(r['checks'].values())]
criterion_failures=sum(not value for r in requests for value in r['checks'].values())
report=objective(len(requests),len(failed),98)
```

```
assert report['successes']==195 and report['allowed_failures']==4
assert not report['meets'] and report['remaining_budget']==-1
assert len(failed)==5 and criterion_failures==7  # overlapping checks count once/request
assert requests[9]['kind']=='correct_refusal' and all(requests[9]['checks'].values())
print('Window manifest:',WINDOW)
print('Constructed window W-A: 200 eligible requests; 3 separate health checks excluded by
the declared rule.')
print('Failed request IDs:',[r['id'] for r in failed],'failed
criteria:',criterion_failures)
print('Count-based objective:',report)
print('Excluded records:',excluded)
# Separate pictured window and a counterfactual copy, not a relabelled original.
picture_original=objective(100,3,98)
picture_counterfactual=objective(100,2,98)
assert not picture_original['meets'] and picture_counterfactual['meets']
print('Picture original:',picture_original)
print('Separate counterfactual copy:',picture_counterfactual)
assert picture_original['failures']==3  # original history is preserved
assert objective(12,0,98)['allowed_failures']==0
assert not objective(12,1,98)['meets']
print('At 98% over 12 requests, the maximum integer failure count is 0, not a rounded-up
fraction.')
print('Synthetic check flags only: no actual reliability interval, timing measurement or
future guarantee.')
```

Expected counts,history preservation and limits

1. Predict main success count, allowed failures and objective decision before running or tracing. The program constructs 200 request fixtures with 5 failed requests but 7 failed individual checks, then counts each failed request once. It includes a correctly handled refusal among successes and reports 3 separate health-check exclusions. Exact integer arithmetic avoids rounding a fractional allowed-failure count upward.

   The original and counterfactual picture summaries are separate objects; the original stays 97/3. All flags and window assignments are synthetic. No request timing, live reliability or actual retry improvement is measured by this code.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: evaluate a fresh objective and request count

> Use the new percentage and denominator before judging the result.

Complete independent 95%/80-request task

Objective 95% over 80 **eligible request**s. Five fail. Compute allowed failures and observed success.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Objective 95% over 80 **eligible request**s. Five fail. Compute allowed failures and observed success.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the objective contract with window-level outcomes

> Preserve eligibility rules,failed IDs,exclusions and original histories.

Project task and artifact

Add a labelled **failure budget**s and **service objective**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **failure budget**s and **service objective**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a bounded interval claim and preserved history

> A met target does not establish zero incidents or the next window’s outcome.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **failure budget**s and **service objective**s

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Service objective	A declared measured target over a specified scope
Failure budget	The failures allowed under that target
Eligible request	A request included under the declared denominator rule

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: Meeting one interval objective does not establish every future interval or zero critical failures.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the service reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Failure budgets and service objectives — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
In the interval 200 requests are eligible and 5 fail.	For a fictional objective at least 98% of 200 eligible requests must pass all contract checks.	195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: A reliability allowance. A practice service requires at least 98 successes in each window of 100 **eligible request**s.

By the end, I can	How I will show it
Explain the mechanism and terminology of failure budgets and service objectives	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A reliability allowance. A practice service requires at least 98 successes in each window of 100 **eligible request**s.
- Which labelled step matches this detail? Three failures misses the target by one request.
- 1. Define success
- 2. Compute allowance
- 3. Compare record
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare record Three failures misses the target by one request. An objective needs a measurement window, denominator and success definition.

An honest refusal can be success under an agreed contract.

Changed-case question and optional hint

- Changed-case practice: Counterfactual: change one failed outcome to a success in a separate copy of this 100-request window. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. 98 successes and 2 failures meet the stated window target.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: 98 successes and 2 failures meet the stated window target. The counterfactual copy changes one outcome while its denominator stays 100; preserve the original 97-success/3-failure window.

Repair and retain the record

Review Evaluation pipelines (ai-6/10). Goal: Explain the **mechanism** and terminology of evaluation pipelines. Short practice: Calculate A's and B's correct rates on all four cases.

Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.. Return to this week and explain what you changed.

Review Testing model-dependent behaviour (ai-6/8). Goal: Explain the **mechanism** and terminology of testing model-dependent behaviour.

Short practice: Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?. Return to this week and explain what you changed.

Review Authentication and authorisation (ai-6/5). Goal: Explain the **mechanism** and terminology of authentication and authorisation. Short practice: Explain why Jo's valid volunteer login does not permit editing C.

Which trusted role change allows the action, and why is a role written in the request insufficient?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Failure budgets and service objectives: the case



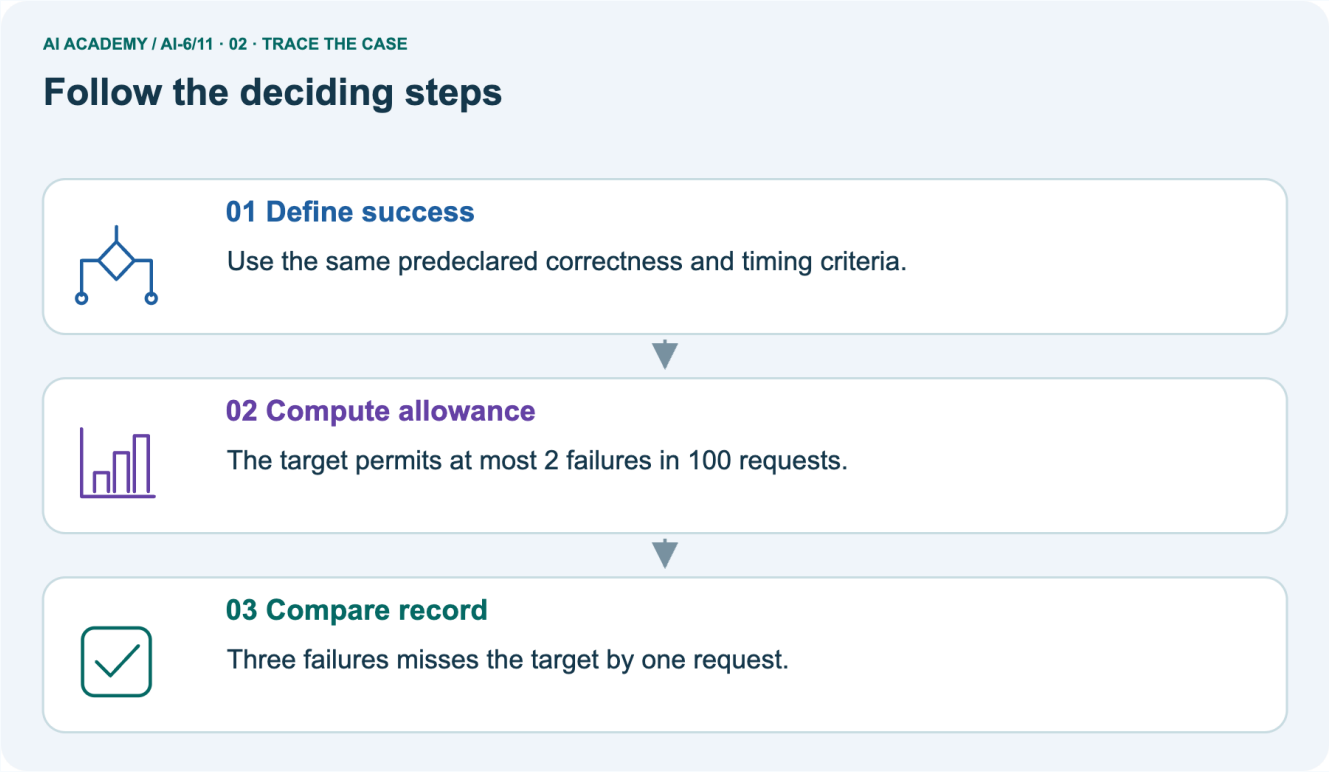
Follow the labelled picture with the matching explanation. Open the vector diagram

A reliability allowance. A practice service requires at least 98 successes in each window of 100 **eligible request**s. Supplied facts: Current window has 97 successes and 3 failures.

Success includes a correct refusal within the agreed time limit.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

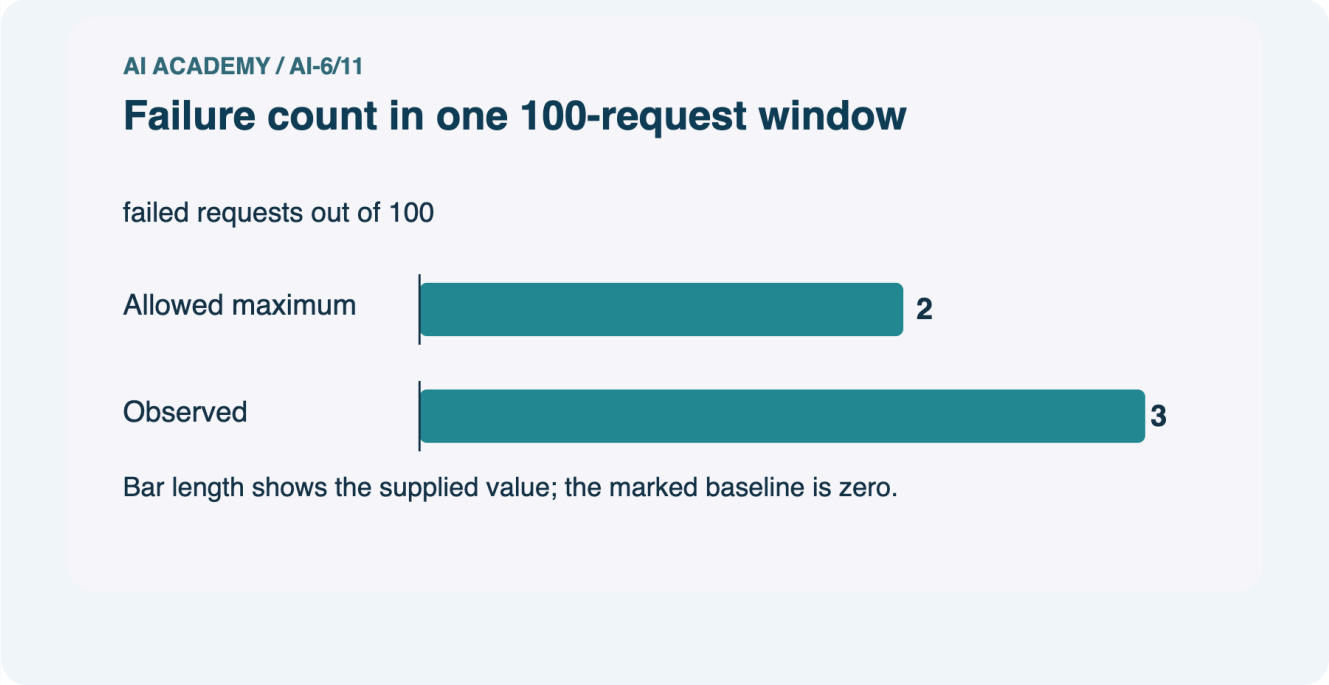


Follow the labelled picture with the matching explanation. Open the vector diagram

Define success: Use the same predeclared correctness and timing criteria. → Compute allowance: The target permits at most 2 failures in 100 requests. → Compare record: Three failures misses the target by one request.

Point to each numbered stage and explain which recorded **input** it uses.

03. Failure count in one 100-request window

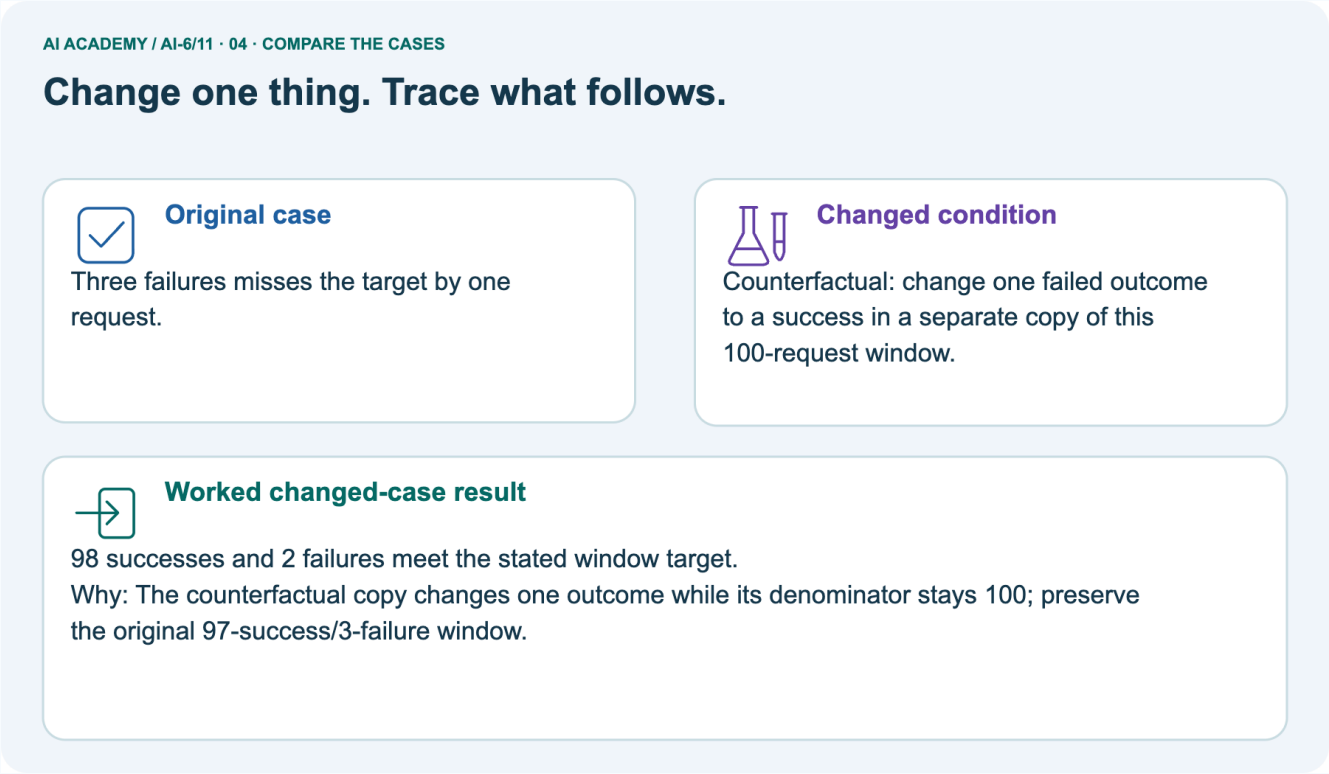


Follow the labelled picture with the matching explanation. Open the vector diagram

Failure count in one 100-request window. Units: failed requests out of 100. Allowed maximum: 2. Observed: 3.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.

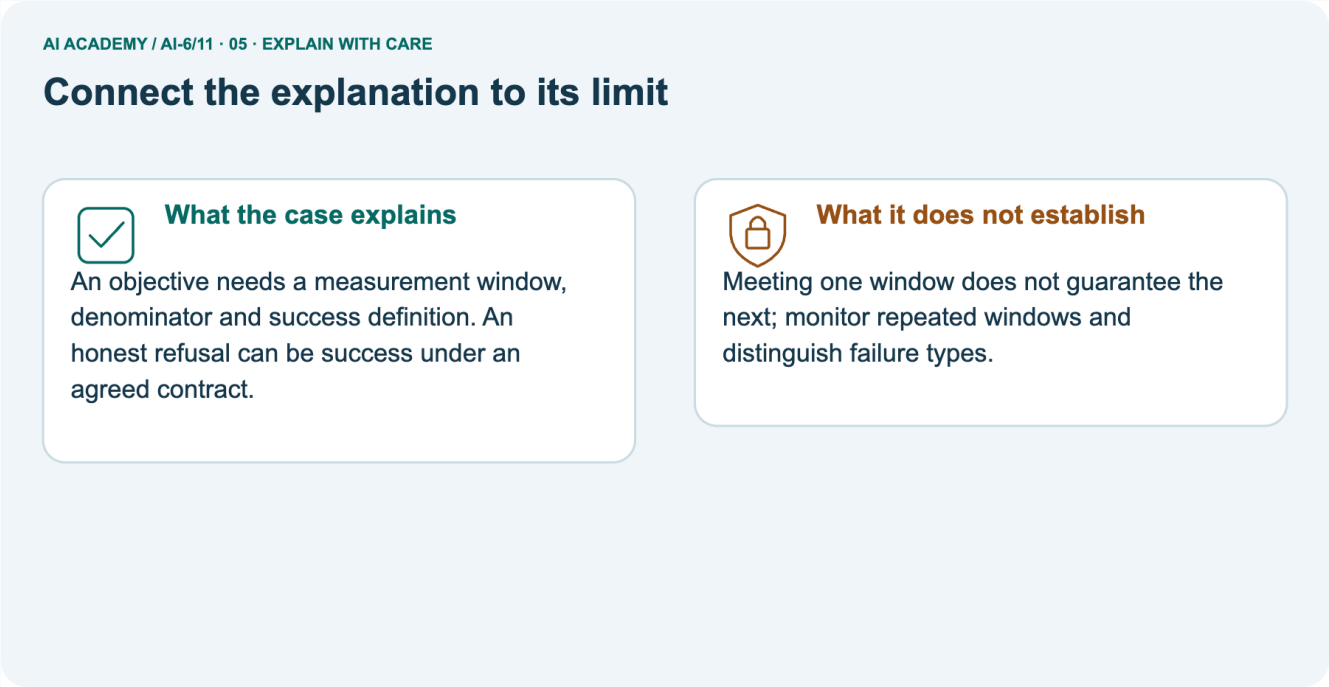


Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Three failures misses the target by one request. Changed condition: Counterfactual: change one failed outcome to a success in a separate copy of this 100-request window. Expected result: 98 successes and 2 failures meet the stated window target. Reason: The counterfactual copy changes one outcome while its denominator stays 100; preserve the original 97-success/3-failure window.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



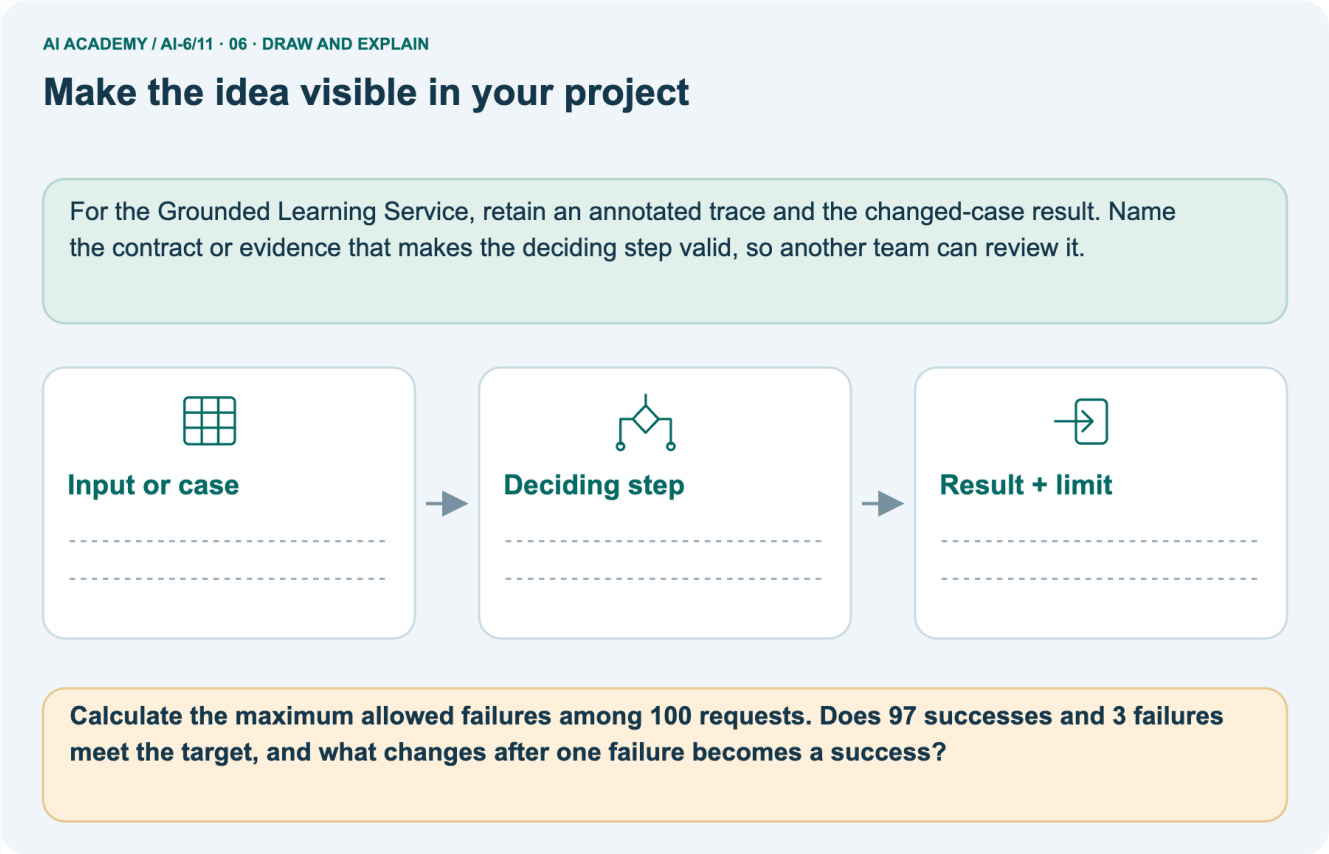
Follow the labelled picture with the matching explanation. Open the vector diagram

An objective needs a measurement window, denominator and success definition. An honest refusal can be success under an agreed contract.

Limit: Meeting one window does not guarantee the next; monitor repeated windows and distinguish failure types.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the maximum allowed failures among 100 requests.

Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Failure budgets and service objectives: the case

02. Follow the deciding steps

03. Failure count in one 100-request window

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.